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Voice-activated ambulance booking

Abstract

Embodiments of the present disclosure relate to systems, methods, and user interfaces for providing voice-activated ambulance booking. Initially, a non-medical user initiates a workflow, via a voice command, that enables the user to provide information about a patient that can be utilized to prioritize the booking of ambulances for patients. Upon providing the initial voice command, the workflow is provided, via a user interface of the user device. The workflow comprises a series of questions corresponding to a status of the patient that can easily be answered merely by observing the patient (e.g., is the patient conscious, is the patient breathing, is the patient bleeding). Based on responses provided by the user, an ambulance is requested for the patient. User device vitals including battery charge levels and signal strength can be used to resolve conflicts if multiple requests are made.

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Background/Summary

INCORPORATION BY REFERENCE; DISCLAIMER

(1) The following application is hereby incorporated by reference: application Ser. No. 16/127,609 filed on Sep. 11, 2018. The Applicant hereby rescinds any disclaimer of claim scope in the parent

application(s) or the prosecution history thereof and advises the USPTO that the claims in this application may be broader than any claim in the parent application(s).

BACKGROUND

(2) Medical emergencies often demand immediate lifesaving interventions or time-sensitive treatments. Generally, finding an ambulance involves calling an operational center (e.g., **911**) and providing an address. In turn, the operational center may connect the caller to the most appropriate ambulance based on the location and availability of ambulances. Current systems allow the user to request the closest available ambulance by using geo-location information provided by a user device (e.g., a mobile phone). In some instances, however, the user device may be locked and the caller may not be able to operate the device. In other instances, the user may be injured and unable to operate the device. Even when a caller is able to call the operational center, the user may fail to properly convey the status of the patient or the user device that may affect prioritization of the call and allocation of resources (e.g., the ambulance).

BRIEF SUMMARY

(3) This summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. This summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used as an aid in determining the scope of the claimed subject matter.

(4) Embodiments of the present disclosure relate to systems, methods, and user interfaces for providing voice-activated ambulance booking. More particularly, embodiments of the present disclosure empower a non-medical user to initiate a workflow, via a voice command, that enables the user to provide information about a patient that can be utilized to prioritize the booking of ambulances for patients. In embodiments, the voice command may initiate the workflow even if a user device is locked. Upon providing the initial voice command, the workflow is provided, via a user interface of the user device. The workflow comprises a series of questions corresponding to a status of the patient that can easily be answered merely by observing the patient (e.g., is the patient conscious, is the patient breathing, is the patient bleeding). Based on responses provided by the user, an ambulance is requested for the patient. In embodiments, user device vitals including battery charge levels and signal strength can be used to resolve conflicts if multiple requests are made. In some embodiments, the workflow can be initiated even if the user device is in an offline mode. Moreover, the workflow can be integrated with other ambulance booking services and/or into an electronic medical record (EMR) of the patient.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

(1) FIG. 1 is a block diagram of an exemplary operating environment suitable to implement embodiments of the present invention;

(2) FIG. 2 depicts an exemplary framework of a voice-activated ambulance booking system suitable to implement embodiments of the present invention;

(3) FIG. 3 depicts a flow diagram illustrating the workflow and prioritization for multiple requests from multiple users of the voice-activated ambulance booking system, in accordance with an embodiment of the present invention;

(4) FIG. 4 depicts a flow diagram illustrating the workflow for a single request from a single user of the voice-activated ambulance booking system, in accordance with an embodiment of the present invention; and

(5) FIG. 5 depicts a flow diagram illustrating voice-activated ambulance booking, in accordance with embodiments of the present invention.

DETAILED DESCRIPTION

(6) The subject matter of the present invention is described with specificity herein to meet statutory requirements. However, the description itself is not intended to limit the scope of this patent. Rather, the inventors have contemplated that the claimed subject matter might also be embodied in other ways, to include different steps or combinations of steps similar to the ones described in this document, in conjunction with other present or future technologies. Moreover, although the terms “step” and/or “block” might be used herein to connote different elements of methods employed, the terms should not be interpreted as implying any particular order among or between various steps herein disclosed unless and except when the order of individual steps is explicitly stated.

(7) As noted in the background, medical emergencies often demand immediate lifesaving interventions or time-sensitive treatments. Generally, finding an ambulance involves calling an operational center (e.g., 911) and providing an address. In turn, the operational center may connect the caller to the most appropriate ambulance based on the location and availability of ambulances. Current systems allow the user to request the closest available ambulance by using geo-location information provided by a user device (e.g., a mobile phone). In some instances, however, the user device may be locked and the caller may not be able to operate the device. In other instances, the user may be injured and unable to operate the device. Even when a caller is able to call the operational center, the user may fail to properly convey the status of the patient or the user device that may affect prioritization of the call and allocation of resources (e.g., the ambulance).

(8) Embodiments of the present disclosure relate to systems, methods, and user interfaces for providing voice-activated ambulance booking. More particularly, embodiments of the present disclosure empower a non-medical user to initiate a workflow, via a voice command, that enables the user to provide information about a patient that can be utilized to prioritize the booking of ambulances for patients. In embodiments, the voice command may initiate the workflow even if a user device is locked. Upon providing the initial voice command, the workflow is provided, via a user interface of the user device. The workflow comprises a series of questions corresponding to a status of the patient that can easily be answered merely by observing the patient (e.g., is the patient conscious, is the patient breathing, is the patient bleeding). Based on responses provided by the user, an ambulance is requested for the patient. In embodiments, mobile device vitals including battery charge levels, signal strength, and the location of the user device from the nearest hospital can be used to resolve conflicts if multiple requests are made. In some embodiments, the workflow can be initiated even if the user device is in an offline mode. Moreover, the workflow can be integrated with other ambulance booking services and/or into an electronic medical record (EMR) of the patient.

(9) Accordingly, one embodiment of the present disclosure is directed to one or more computer storage media having computer-executable instructions embodied thereon that, when executed by a computer, causes the computer to perform operations for providing voice-activated ambulance booking. The operations include receiving, at a user device, a voice command from a user. The voice command initiates a workflow that enables the user to provide information about a patient to a booking engine. The operations also include providing the workflow to the user, via a user interface of the user device. The workflow comprises a series of questions corresponding to a status of the patient. The operations further include requesting an ambulance for the patient. The ambulance is requested based on responses from the user, via the user interface of the user device, to the series of questions, and a detected location of the user device.

(10) In another embodiment, the present disclosure directed to a computerized method providing voice-activated ambulance booking. The method comprises receiving, at a user device, a voice command from a user. The voice command initiates a workflow that enables the user to provide information about a patient to a booking engine. The method also comprises providing the workflow to the user, via a user interface of the user device. The workflow comprises a series of questions corresponding to a status of the patient. The method further comprises requesting an ambulance for the patient. The ambulance is requested based on responses from the user, via the

user interface of the user device, to the series of questions, and a detected location of the user device. The method also comprises identifying the ambulance for the patient that is available. The method further comprises determining a plurality of requests have been received from other patients. The method also comprises prioritizing requests from the patient and the other patients based on responses of the patient and the other patients to the series of questions.

(11) In yet another embodiment, the present disclosure is directed to a system for providing voice-activated ambulance booking. The system comprises a processor; and a computer storage medium storing computer-usable instructions that, when used by the processor, cause the processor to: receive, at a user device, a voice command from a user, the voice command initiating a workflow that enables the user to provide information about a patient to a booking engine; provide the workflow to the user, via a user interface of the user device, the workflow comprising a series of questions corresponding to a status of the patient; request an ambulance for the patient, the ambulance requested based on responses from the user, via the user interface of the user device, to the series of questions, and a detected location of the user device; and identify the ambulance for the patient that is available.

(12) Having briefly described embodiments of the present invention, an exemplary operating environment suitable for use in implementing embodiments of the present invention is described below. FIG. 1 provides an aspect of an example operating environment with which embodiments of the present invention may be implemented. The aspect of an operating environment is illustrated and designated generally as reference numeral **100**.

(13) Example operating environment **100** comprises a general purpose computing device in the form of a control server **102**. Exemplary components of the control server **102** comprise a processing unit, internal system memory, and a suitable system bus for coupling various system components, including database cluster **104**, with the control server **102**. The system bus might be any of several types of bus structures, including a memory bus or memory controller, a peripheral bus, and a local bus, using any of a variety of bus architectures. Exemplary architectures comprise Industry Standard Architecture (ISA) bus, Micro Channel Architecture (MCA) bus, Enhanced ISA (EISA) bus, Video Electronic Standards Association (VESA) local bus, and Peripheral Component Interconnect (PCI) bus, also known as Mezzanine bus.

(14) Control server **102** typically includes therein, or has access to, a variety of computer-readable media, for instance, database cluster **104**. Computer-readable media can be any available media that might be accessed by control server **102**, and includes volatile and nonvolatile media, as well as, removable and nonremovable media. Computer-readable media might include computer storage media. Computer storage media includes volatile and nonvolatile media, as well as removable and nonremovable media implemented in any method or technology for storage of information, such as computer-readable instructions, data structures, program modules, or other data. In this regard, computer storage media might comprise RAM, ROM, EEPROM, flash memory or other memory technology, CD-ROM, digital versatile disks (DVDs) or other optical disk storage, magnetic cassettes, magnetic tape, magnetic disk storage, or other magnetic storage device, or any other medium which can be used to store the desired information and which may be accessed by the control server **102**. Computer storage media does not comprise signals per se. Combinations of any of the above also may be included within the scope of computer-readable media.

(15) The computer storage media discussed above and illustrated in FIG. 1, including database cluster **104**, provide storage of computer-readable instructions, data structures, program modules, and other data for the control server **102**. In some embodiments, data cluster **104** takes the form of a cloud-based data store, and in some embodiments is accessible by a cloud-based computing platform.

(16) The control server **102** might operate in a computer network **106** using logical connections to one or more remote computers **108**. Remote computers **108** might be located at a variety of locations in a medical or research environment, including clinical laboratories (e.g., molecular

diagnostic laboratories), hospitals and other inpatient settings, veterinary environments, ambulatory settings, medical billing and financial offices, hospital administration settings, home healthcare environments, and providers' offices. Providers may comprise a treating physician or physicians; specialists such as surgeons, radiologists, cardiologists, and oncologists; emergency medical technicians; physicians' assistants; nurse practitioners; nurses; nurses' aides; pharmacists; dietitians; microbiologists; laboratory experts; laboratory technologists; genetic counselors; researchers; veterinarians; students; and the like.

(17) The remote computers **108** might also be physically located in nontraditional medical care environments so that the entire healthcare community might be capable of integration on the network. The remote computers **108** might be personal computers, servers, routers, network PCs, peer devices, other common network nodes, or the like and might comprise some or all of the elements described above in relation to the control server **102**. The devices can be personal digital assistants or other like devices.

(18) Exemplary computer networks **106** comprise local area networks (LANs) and/or wide area networks (WANs). Such networking environments are commonplace in offices, enterprise-wide computer networks, intranets, and the Internet. When utilized in a WAN networking environment, the control server **102** might comprise a modem or other means for establishing communications over the WAN, such as the Internet. In a networked environment, program modules or portions thereof might be stored in association with the control server **102**, the database cluster **104**, or any of the remote computers **108**. For example, various application programs may reside on the memory associated with any one or more of the remote computers **108**. It will be appreciated by those of ordinary skill in the art that the network connections shown are exemplary and other means of establishing a communications link between the computers (e.g., control server **102** and remote computers **108**) might be utilized.

(19) In operation, an organization might enter commands and information into the control server **102** or convey the commands and information to the control server **102** via one or more of the remote computers **108** through input devices, such as a keyboard, a pointing device (commonly referred to as a mouse), a trackball, or a touch pad. Other input devices comprise microphones, satellite dishes, scanners, or the like. Commands and information might also be sent directly from a remote healthcare device to the control server **102**. In addition to a monitor, the control server **102** and/or remote computers **108** might comprise other peripheral output devices, such as speakers and a printer.

(20) In some embodiments, control server **102** is a computing system or platform made up of one or more computing devices. Embodiments of control server **102** may be a distributed computing system, a centralized computing system, a single computer such as a desktop or laptop computer or a networked computing system. Thus, in some embodiments, control server **102** comprises a multi-agent computer system with software agents.

(21) Turning now to FIG. 2, an exemplary framework of a voice-activated ambulance booking system **200** is shown, in accordance with an aspect of the present invention. It should be understood that this and other arrangements described herein are set forth only as examples. Other arrangements and elements (e.g., machines, interfaces, functions, orders, and groupings of functions, etc.) can be used in addition to or instead of those shown, and some elements may be omitted altogether. Further, many of the elements described herein are functional entities that may be implemented as discrete or distributed components or in conjunction with other components, and in any suitable combination and location. Various functions described herein as being performed by one or more entities may be carried out by hardware, firmware, and/or software. For instance, various functions may be carried out by a processor executing instructions stored in memory. The voice-activated ambulance booking system **200** may be implemented via any type of computing device, such as computing device **100** described above with reference to FIG. 1, for example.

(22) The voice-activated ambulance booking system **200** generally operates to provide voice-activated ambulance booking. More particularly, the voice-activated ambulance booking system **200** empowers a non-medical user to initiate a workflow, via a voice command, that enables the user to provide information about a patient that can be utilized to prioritize the booking of ambulances for patients. Upon a user providing the initial voice command, the workflow is provided by the voice-activated ambulance booking system **200**, via a user interface of the user device. The workflow comprises a series of questions corresponding to a status of the patient that can easily be answered merely by observing the patient (e.g., is the patient conscious, is the patient breathing, is the patient bleeding). Based on responses provided by the user, an ambulance is requested for the patient.

(23) As shown in FIG. 2, the voice-activated ambulance booking system **200** includes, among other components not shown, database(s) **204a-204n**, ambulance(s) **206a-206n**, user device(s) **208a-208n**, and booking engine **210**. It should be understood that the voice-activated ambulance booking system **200** shown in FIG. 2 is an example of one suitable computing system architecture. Each of the components shown in FIG. 2 may be implemented via any type of computing device, such as computing device **100** described with reference to FIG. 1, for example.

(24) The components may communicate with each other via a network **202**, which may include, without limitation, one or more local area networks (LANs) and/or wide area networks (WANs). Such networking environments are commonplace in offices, enterprise-wide computer networks, intranets, and the Internet. It should be understood that any number of booking engines may be employed within the voice-activated ambulance booking system **200** within the scope of the present disclosure. Each may comprise a single device or multiple devices cooperating in a distributed environment. For instance, the booking engine **210** (or any of its components: voice command component **212**, workflow component **214**, prioritization component **216**, identification component **220**) may be provided via multiple devices arranged in a distributed environment that collectively provide the functionality described herein. In other embodiments, a single device may provide the functionality of multiple components of the voice-activated ambulance booking system **200**. For example, a single device may provide the booking engine **210** and/or the database(s) **204a-204n**. In some embodiments, some or all functionality provided by the booking engine **210** (or any of its components) and/or the database(s) **204a-204n** may be provided by a user device. Additionally, other components not shown may also be included within the network environment.

(25) Generally, the booking engine **210** enables a user to initiate a workflow on a user device, by voice-command, that enables the user to request and book an ambulance for a patient. In embodiments, the voice command may initiate the workflow even if a user device is locked. Upon providing the initial voice command, the workflow is provided, via a user interface of the user device.

(26) Although the workflow is described herein as being provided via a user interface, it is contemplated that the user interface may not be limited to a visual workflow. For example, the workflow may be an interaction between voice commands received from a user and voice responses provided via the user device by the application. The workflow may also be an interaction between a combination of voice inputs and outputs and visual inputs and outputs. The voice commands and voice responses enable the application to be utilized even when the user device is in a locked mode. In some embodiments, the voice command that initiates the workflow when the user device is locked enables the remainder of the workflow via voice inputs and outputs or visual inputs and outputs, or a combination of both. In this way, the initial voice command serves to unlock the user device for the purpose of utilizing the workflow.

(27) The workflow comprises a series of questions corresponding to a status of the patient that can easily be answered merely by observing the patient (e.g., is the patient conscious, is the patient breathing, is the patient bleeding). Based on responses provided by the user, an ambulance is requested for the patient. In embodiments, mobile device vitals including battery charge levels,

signal strength, and the location of the user device from the nearest hospital can be used to resolve conflicts if multiple requests are made. In some embodiments, the workflow can be initiated even if the user device is in an offline mode. Moreover, the workflow can be integrated with other ambulance booking services and/or into an electronic medical record (EMR) of the patient.

(28) As described above, the booking engine **210** includes several components including voice command component **212**, workflow component **214**, prioritization component **216**, and identification component **220**.

(29) The voice command component **212** generally receives, from a user device, a voice command from a person. The voice command initiates a workflow that enables the person to provide information about a patient to a booking engine.

(30) The workflow component **214** generally provides the workflow to the user, via a user interface of the user device. The workflow comprises a series of questions corresponding to a status of the patient. For example, the workflow component **214** may prompt the user, via the user interface of the user device, to provide information corresponding to the consciousness, breathing, and/or bleeding status of the patient.

(31) The prioritization component **216** generally requests an ambulance for the patient. The ambulance is requested based on responses from the user, via the user interface of the user device, to the series of questions, and a detected location of the user device.

(32) In an embodiment, the prioritization component **216** determines a plurality of ambulances are available for a single user request. Ambulances that have an outstanding request from another patient received prior to the current request from the patient are eliminated from consideration by the prioritization component **216**.

(33) In another embodiment, the prioritization component **216** determines a plurality of requests have been received from other patients and only a single ambulance is available. In this case, the prioritization component **216** prioritizes requests from the patient and the other patients based on responses to the series of questions. As described, the series of questions may comprise breathing status, bleeding status, and/or consciousness status of the patient and the other patients.

(34) Additionally, or alternatively, the prioritization component **216** further prioritizes requests based on battery status of devices making the requests, signal strength of devices making the requests, and/or detected location of the devices.

(35) The identification component **218** generally identifies the ambulance for the patient that is available, best suited, or based on priority information received from prioritization component **216**. In some embodiments, the identification component **218** communicates an offer corresponding to the user. The offer may include an estimated time of arrival, estimated costs, insurance coverage information, and the like. In some embodiments, an acknowledgement may be received from the user indicating the user has accepted the offer. The voice-activated ambulance booking may then be confirmed by the identification component **218** to all involved parties. In some embodiments, the identification component **218** enables the user to track the ambulance in real-time, via the user interface, until it reaches the detected location of the user device. Additionally, information of the patient may be communicated by the identification component **218** to an ambulance device, or an EMR that may be accessible by an ambulance device. An emergency contact of the patient may also be notified by the identification component **218**. The emergency contact may be provided by the patient at the time of registration with the booking engine **210**.

(36) Referring now to FIG. 3, an exemplary flow diagram **300** illustrates the workflow and prioritization for multiple requests from multiple users of the voice-activated ambulance booking system, in accordance with an embodiment of the present invention. Method **300** may be performed by any computing device (such as computing device described with respect to FIG. 1) with access to voice-activated ambulance booking system (such as the one described with respect to FIG. 2) or by one or more components of the voice-activated ambulance booking system.

(37) Initially, as shown at step **302**, a user interface prompts a user to indicate the consciousness of

the patient. For example, the user may be prompted to indicate whether the patient is unconscious, sub-conscious, or conscious. If the user selects the patient is conscious, the user interface prompts the user, at step **304** to indicate whether the patient is breathing normal. If the user selects the patient is breathing normal or not, or if the user selects the patient is unconscious or sub-conscious in the previous step, the user interface prompts the user to indicate whether the patient is bleeding, at step **306**. After the user selects whether the patient is bleeding, at step **306**, the responses are provided to the voice-activated ambulance booking system, at step **310**, for prioritization. Once a patient and ambulance is selected by the voice-activated ambulance booking system, the selected ambulance is assigned to the selected patient, at step **320**. Information provided by the user via the user interface may be provided to the selected ambulance so the clinicians are aware of the patient situation. Similarly, the information may also be provided to the health care facility that the patient is transported to by the ambulance.

(38) In FIG. **4**, an exemplary flow diagram **400** illustrates the workflow for a single request from a single user of the voice-activated ambulance booking system, in accordance with an embodiment of the present invention. Method **400** may be performed by any computing device (such as computing device described with respect to FIG. **1**) with access to voice-activated ambulance booking system (such as the one described with respect to FIG. **2**) or by one or more components of the voice-activated ambulance booking system.

(39) Initially, as shown at step **402**, a user interface prompts a user to indicate the consciousness of the patient. For example, the user may be prompted to indicate whether the patient is unconscious, sub-conscious, or conscious. If the user selects the patient is conscious, the user interface prompts the user, at step **404** to indicate whether the patient is breathing normal. If the user selects the patient is breathing normal or not, or if the user selects the patient is unconscious or sub-conscious in the previous step, the user interface prompts the user to indicate whether the patient is bleeding, at step **406**. After the user selects whether the patient is bleeding, at step **406**, the responses are provided to the voice-activated ambulance booking system, and an ambulance is selected by the voice-activated ambulance booking system and assigned to the patient, at step **420**.

(40) Turning now to FIG. **5**, a flow diagram is provided illustrating a method **500** for voice-activated ambulance booking, in accordance with embodiments of the present invention. Method **500** may be performed by any computing device (such as computing device described with respect to FIG. **1**) with access to voice-activated ambulance booking system (such as the one described with respect to FIG. **2**) or by one or more components of the voice-activated ambulance booking system.

(41) Initially, at step **502**, a person registers with the voice-activated ambulance booking system (such as via an online service or application). The person provides important information that is saved in a central database. The important information may include name, address, phone number, age, gender, and the like.

(42) Upon determining an ambulance is needed, a user (e.g., patient or patient's attendant) may initiate or log-in to the voice-activated ambulance booking system, at step **504**, with a voice command or emergency call feature via a user device (e.g., a mobile device).

(43) At step **506**, the user submits a request for an ambulance through a voice command or manually, via a user interface provided by the voice-activated ambulance booking system (i.e., the client). The user interface provides questions to the user regarding consciousness, breathing, and bleeding status for the patient. Additionally, or alternatively, the questions may be provided audibly to the user. The user may provide answers to the questions via the user interface, or by voice command.

(44) A user corresponding to the user interface provided by the voice-activated ambulance booking system (i.e., the client) enters a request state, at step **508**.

(45) Based on network availability, the booking process may work in an online or offline mode, at step **510**. For example, in an offline mode, the request may be sent via text message. The booking

process enables the request state to be communicated to the voice-activated ambulance booking system.

(46) Once the request for the ambulance enters the network, such as through a gateway, the user interface provided by the voice-activated ambulance booking system (i.e., the client) enters a waiting state, at step **512**.

(47) An available ambulance may be identified, at step **514**. Several factors may be utilized to determine the availability of an ambulance, including the distance between the ambulance and the patient, traffic density, and the like.

(48) In embodiments, at step **516**, it may be determined whether a single ambulance was identified for a single client (i.e., patient) or multiple clients (i.e., patients) or whether multiple ambulances were identified for a single client.

(49) It is determined, at step **518**, if a single ambulance was identified for multiple clients or multiple ambulances were identified for a single client.

(50) In a single ambulance identified for multiple clients embodiment, the voice-activated ambulance booking system utilizes the breathing, bleeding, and consciousness status of the patients, at step **520** to prioritize patients in the same locality.

(51) Other factors corresponding to the user device may be utilized, at step **522**, if there is more than one patient with the same priority. For example, the user device with the lowest remaining charge may be given priority so the patient can be located before the user device runs out of battery power. Similarly, the user device with the weakest signal strength may be given priority so the patient can be located before the signal drops or fails. Alternatively, the patient that is the farthest from the hospital may be given priority so the patient can be transported to the hospital while an alternative ambulance is identified for the patient that is closer to the hospital.

(52) At step **524**, the ambulance is assigned to the best suited patient (i.e., the patient that has the highest priority). Other patients may need to resend a request for an ambulance via voice commands or manually via the user interface.

(53) In a multiple ambulance for a single client embodiment, all ambulances with requests from other patients are eliminated, at step **530**, if the requests were made prior to the current request.

(54) At step **532**, the best suited ambulance is assigned to the patient. All other ambulances remain in or enter an available status.

(55) In each embodiment (a single ambulance identified for multiple clients embodiment and a multiple ambulance for a single client embodiment) or if a single ambulance is identified for a single client, the ambulance sends an offer, along with an estimated time of arrival, to the user, at step **526**.

(56) The ambulance then enters a waiting status, at step **528**. If the user acknowledges the offer, at step **534**, before a timeout (i.e., a configurable period of time the user must accept the offer in before the offer expires), at step **536**, the booking is confirmed, at step **542**, and the ambulance remains in a busy status.

(57) The user may track, at step **544**, the ambulance in real-time, via the user interface of the voice-activated ambulance booking system, until it reaches its destination. Once the patient enters the ambulance, at step **546**, the user is prompted to determine if the user and the patient are the same person. If the user and the patient are determined to be the same person, at step **548**, the voice-activated ambulance booking system requests information corresponding to the patient from the database, at step **550**. If the user and the patient are not determined to be the same person, at step **550**, a missed call from the RMN of the patient is provided, at step **552**, and the voice-activated ambulance booking system requests information corresponding to the patient from the database. In each case, the patient is transported to the hospital and friends and/or family of the patient is notified, at step **554**.

(58) If the user acknowledges the request, at step **534**, but does not do so before a timeout, at step **536**, the request expires, at step **538**, and the ambulance enters an available status, at step **540**,

where it can be identified for a single patient or multiple patients, at step 516. If the user does not acknowledge the request, at step 534, the request expires, at step 538, and the ambulance enters an available status, at step 540, where it can be identified for a single patient or multiple patients, at step 516.

(59) In practice, a request R1 may be received via text message (i.e., offline). A second request R2 may be received via an application (i.e., online). In cases where each request is prioritized the same, a tiebreaker may be used. Assume signal strength and batter percentage are low for the device sending request R1 as compared to the device sending request R2, request R1 receives higher priority and acceptance and acknowledgement of request R1 will be provided to user via text message.

(60) In another example, a request R2 may be received via an application (i.e., online). A second request R2 may be received via text message (i.e., offline). In cases where each request is prioritized the same, a tiebreaker may be used. Assume signal strength and batter percentage are low for the device sending request R2 as compared to the device sending request R1, request R2 receives higher priority and acceptance and acknowledgement of request R2 will be provided to user via the application.

(61) Many different arrangements of the various components depicted, as well as components not shown, are possible without departing from the spirit and scope of the present invention.

Embodiments of the present invention have been described with the intent to be illustrative rather than restrictive. Alternative embodiments will become apparent to those skilled in the art that do not depart from its scope. A skilled artisan may develop alternative means of implementing the aforementioned improvements without departing from the scope of the present invention.

(62) It will be understood that certain features and subcombinations are of utility and may be employed without reference to other features and subcombinations and are contemplated within the scope of the claims. Not all steps listed in the various figures need be carried out in the specific order described. Accordingly, the scope of the invention is intended to be limited only by the following claims.

Claims

1. One or more non-transitory computer-readable media storing instructions which, when executed by one or more hardware processors, cause performance of operations comprising: presenting, at a first device of a plurality of source devices while the first device is in a locked state, a workflow comprising one or more prompts for gathering information about a patient; generating, at the first device while the first device is in the locked state, a first ambulance request based at least in part on one or more user responses to the one or more prompts; receiving, over a network by an ambulance booking system, a plurality of ambulance requests from a respective plurality of source devices, the plurality of ambulance requests comprising the first ambulance request received from the first device of the plurality of source devices; obtaining, by the ambulance booking system, device information for each of the plurality of source devices; wherein the device information, for the first device of the plurality of sources devices, comprises a detected location of the first device and one or more of: a current battery level of the first device or a current signal strength of the first device; assigning, by the ambulance booking system, a level of priority to each particular ambulance request of the plurality of ambulance requests based at least in part on the device information for the respective source device corresponding to the particular ambulance request; selecting the first ambulance request of the plurality of ambulance requests for prioritization, based at least on the level of priority assigned to the first ambulance request; booking, by the ambulance booking system, an ambulance in response to the selected first ambulance request; tracking in real time, at the first device, progress of the ambulance toward the detected location of the first device.

2. The one or more non-transitory computer-readable media of claim 1, wherein the one or more

user responses to the one or more prompts comprise one or more of voice input or manual input to the first device.

3. The one or more non-transitory computer-readable media of claim 1, wherein selecting the first ambulance request of the plurality of ambulance requests for prioritization is further based on a patient condition indicated by the one or more user responses to the one or more prompts.

4. The one or more non-transitory computer-readable media of claim 1, the operations further comprising: determining, by the ambulance booking system, availability of the ambulance to service the first ambulance request based at least in part on a distance between the ambulance and the detected location of the first device.

5. The one or more non-transitory computer-readable media of claim 1, wherein the first device transmits the first ambulance request to the ambulance booking system while the first device is in the locked state.

6. The one or more non-transitory computer-readable media of claim 1, wherein fewer ambulances are available to respond to the plurality of ambulance requests than a total number of the plurality of ambulance requests.

7. A system comprising: one or more hardware processors; one or more non-transitory computer-readable media; and program instructions stored on the one or more non-transitory computer readable media which, when executed by the one or more hardware processors, cause the system to perform operations comprising: presenting, at a first device of a plurality of source devices while the first device is in a locked state, a workflow comprising one or more prompts for gathering information about a patient; generating, at the first device while the first device is in the locked state, a first ambulance request based at least in part on one or more user responses to the one or more prompts; receiving, over a network by an ambulance booking system, a plurality of ambulance requests from a respective plurality of source devices, the plurality of ambulance requests comprising the first ambulance request received from the first device of the plurality of source devices; obtaining, by the ambulance booking system, device information for each of the plurality of source devices; wherein the device information, for the first device of the plurality of sources devices, comprises a detected location of the first device and one or more of: a current battery level of the first device or a current signal strength of the first device; assigning, by the ambulance booking system, a level of priority to each particular ambulance request of the plurality of ambulance requests based at least in part on the device information for the respective source device corresponding to the particular ambulance request; selecting the first ambulance request of the plurality of ambulance requests for prioritization, based at least on the level of priority assigned to the first ambulance request; booking, by the ambulance booking system, an ambulance in response to the selected first ambulance request; tracking in real time, at the first device, progress of the ambulance toward the detected location of the first device.

8. The system of claim 7, wherein the one or more user responses to the one or more prompts comprise one or more of voice input or manual input to the first device.

9. The system of claim 7, wherein selecting the first ambulance request of the plurality of ambulance requests for prioritization is further based on a patient condition indicated by the one or more user responses to the one or more prompts.

10. The system of claim 7, the operations further comprising: determining, by the ambulance booking system, availability of the ambulance to service the first ambulance request based at least in part on a distance between the ambulance and the detected location of the first device.

11. The system of claim 7, wherein the first device transmits the first ambulance request to the ambulance booking system while the first device is in the locked state.

12. The system of claim 7, wherein fewer ambulances are available to respond to the plurality of ambulance requests than a total number of the plurality of ambulance requests.

13. A method comprising: presenting, at a first device of a plurality of source devices while the first device is in a locked state, a workflow comprising one or more prompts for gathering information

about a patient; generating, at the first device while the first device is in the locked state, a first ambulance request based at least in part on one or more user responses to the one or more prompts; receiving, over a network by an ambulance booking system, a plurality of ambulance requests from a respective plurality of source devices, the plurality of ambulance requests comprising the first ambulance request received from the first device of the plurality of source devices; obtaining, by the ambulance booking system, device information for each of the plurality of source devices; wherein the device information, for the first device of the plurality of sources devices, comprises a detected location of the first device and one or more of: a current battery level of the first device or a current signal strength of the first device; assigning, by the ambulance booking system, a level of priority to each particular ambulance request of the plurality of ambulance requests based at least in part on the device information for the respective source device corresponding to the particular ambulance request; selecting the first ambulance request of the plurality of ambulance requests for prioritization, based at least on the level of priority assigned to the first ambulance request; booking, by the ambulance booking system, an ambulance in response to the selected first ambulance request; tracking in real time, at the first device, progress of the ambulance toward the detected location of the first device; wherein the method is performed by at least one device including a hardware processor.

14. The method of claim 13, wherein the one or more user responses to the one or more prompts comprise one or more of voice input or manual input to the first device.

15. The method of claim 13, wherein selecting the first ambulance request of the plurality of ambulance requests for prioritization is further based on a patient condition indicated by the one or more user responses to the one or more prompts.

16. The method of claim 13, further comprising: determining, by the ambulance booking system, availability of the ambulance to service the first ambulance request based at least in part on a distance between the ambulance and the detected location of the first device.

17. The method of claim 13, wherein the first device transmits the first ambulance request to the ambulance booking system while the first device is in the locked state.
